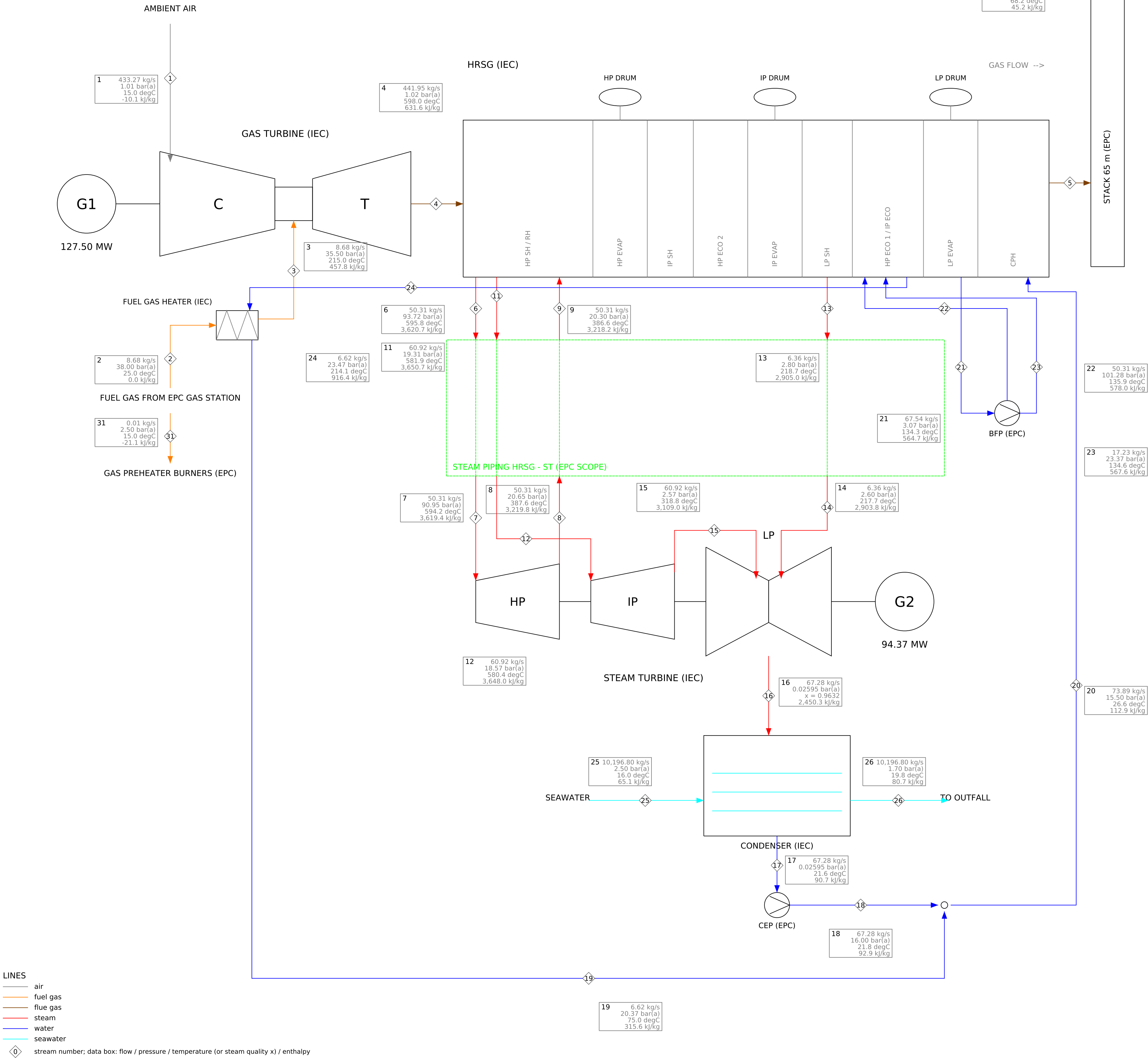




STREAM TABLE - SRC-NG-MEL

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	433.27	1.01	15.0	-10.1	-4.4
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	8.68	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	8.68	35.50	215.0	457.8	4.0
4	GT exhaust gas to HRSG inlet	441.95	1.02	598.0	631.6	279.1
5	Flue gas at HRSG outlet to stack	441.95	1.01	68.2	45.2	20.0
6	HP steam at HRSG outlet (HP SH outlet)	50.31	93.72	595.8	3,620.7	182.2
7	HP steam at ST main stop valve inlet	50.31	90.95	594.2	3,619.4	182.1
8	Cold reheat steam at HP turbine exhaust	50.31	20.65	387.6	3,219.8	162.0
9	Cold reheat steam at HRSG reheater inlet	50.31	20.30	386.6	3,218.2	161.9
10	IP superheated steam (joins cold reheat inside the HRSG)	10.61	20.89	291.3	3,001.3	31.8
11	Hot reheat steam at HRSG outlet	60.92	19.31	581.9	3,650.7	222.4
12	Hot reheat steam at ST IP stop/control valve inlet	60.92	18.57	580.4	3,648.0	222.2
13	LP steam at HRSG outlet	6.36	2.80	218.7	2,905.0	18.5
14	LP steam at ST LP admission	6.36	2.60	217.7	2,903.8	18.5
15	IP turbine exhaust (crossover, internal)	60.92	2.57	318.8	3,109.0	189.4
16	LP turbine exhaust to condenser (UEEP)	67.28	0.0260	x 0.963	2,450.3	164.8
17	Condensate at hotwell outlet / CEP suction	67.28	0.0260	21.6	90.7	6.1
18	Condensate at CEP discharge	67.28	16.00	21.8	92.9	6.3
19	Fuel gas heater water return to condensate	6.62	20.37	75.0	315.6	2.1
20	Condensate to HRSG condensate preheater inlet	73.89	15.50	26.6	112.9	8.3
21	Feedwater from LP drum to boiler feed pumps	67.54	3.07	134.3	564.7	38.1
22	HP feedwater at BFP discharge to HP economiser	50.31	101.28	135.9	578.0	29.1
23	IP feedwater at BFP interstage to IP economiser	17.23	23.37	134.6	567.6	9.8
24	IP economiser water to performance fuel gas heater	6.62	23.47	214.1	916.4	6.1
25	Circulating seawater to condenser inlet	10,196.80	2.50	16.0	68.1	663.9
26	Circulating seawater from condenser outlet	10,196.80	1.70	19.8	80.7	822.8
31	Fuel gas to the station preheater burners (water-bath heaters)	0.01	2.50	15.0	-21.1	-0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

GT output (generator terminals)	127.50	MW
ST output (generator terminals)	94.37	MW
Gross output	221.87	MW
Auxiliary loads	-7.38	MW
Transformer losses (GSU, UAT, MV/LV)	-0.73	MW
NET OUTPUT (HV terminals)	213.77	MW
Heat input (LHV)	412.86	MW
Gross heat rate (LHV)	6,689	kJ/kWh
NET HEAT RATE (LHV)	6,953	kJ/kWh
Net efficiency (LHV)	51.78	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	42.734	Sm3/h
Circulating water flow	36,112	m3/h
GT load	30.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	34.82	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.002 MW (tolerance 0.25 MW)
OK	energy balance COND: residual -0.181 MW (tolerance 0.83 MW)
OK	energy balance FGH: residual +0.002 MW (tolerance 0.25 MW)
OK	energy balance GT: residual -0.019 MW (tolerance 0.41 MW)
OK	energy balance HRSG: residual +1.048 MW vs stated casing loss 1.037 MW (tolerance 0.49 MW)
OK	energy balance PIPE: residual +0.322 MW (tolerance 0.59 MW)
OK	energy balance ST: residual +0.000 MW (tolerance 0.42 MW)
OK	plant load (net, % of the reference case) 34.82 % of plant net output vs PG-11 <= 40.00 (margin +12.95 %)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - SRC-NG-MEL
CASE	Site Reference Conditions, natural gas, GT at minimum emissions-compliant load (
DOCUMENT	ALP-HMB-SRC-NG-MEL
STATUS	preliminary - generated by engine hmb v1.1.1
DATA	b6f4b950c7f3b7e0 2026-09-29
NOTE	all values calculated from the database; do not edit